



How Big Is Big Enough?

The size of the screen that you will need to buy will depend upon multiple attributes of the projector: its native resolution, the minimum and maximum screen size that can be displayed by the projector, luminance, contrast ratio, where you will likely use it, and so on.

An SVGA projector with a luminance rating of 1,000 lumen needs a screen big enough to display the projected image (100 to 200-inch) from a particular distance. Although the projector will display a bigger image if moved back, note that the 1,000-lumen lamp will produce a duller image from a greater distance. After a certain distance, even the best quality screen won't have much light to reflect back. Therefore, make sure that your projector's lumen rating, contrast ratio, and the distance from which you will project, match closely.

The screen size will also depend on the size of the room. Normally, your audience should sit away from the screen at a distance at least twice the image size; any closer, and the image appear as a pixelated mess with visible artefacts. So for an image measuring 100-inch diagonal, the audience should be at least 16 to 18 feet away

than the Epson EMP-S1H, which has a lumen rating of 1,400 but needs only 132 W, making it the most efficient of the LCD lot.

That's not all! The Epson projects very viewable images measuring 330-inch, diagonal from a distance of 11.3 metres, with the 132 W lamp—an impressive feat. We then placed this projector at a distance of three metres from our screen and found that the Epson was bright enough in presentation mode with image size touching 100-inch on our 72x90-inch screen.

As mentioned earlier, DLPs are preferred for movies since their output is generally brighter than LCD projectors. Thus, you will find that most of these projectors come with a selectable aspect ratio of 16:9 for that home theatre movie experience. All projectors in this category offered 16:9, including the four LCD projectors.

Connectors

When it comes to watching movies, connecting your projector to your PC is not always the best option. Standalone DVD players are widely available and you will find yourself using either S-Video or RCA component connectors. Of course, your projector needs to be supportive.

All the projectors, except the Canon LV-S3, either had S-Video or RCA component connectors. Canon just had the D-Sub (for your computer) and a composite connector. Since a composite signal is a mix of colour and sync signals, it is not as good in quality as a component signal (which segregates signals).

Ease Of Use

Setting up a projector is an art! Some struggle to get a projector up, while others are masters at the task and are in demand during meetings and presentations. Of course, the design of the pegs that support the projector body and the ease with which they can be adjusted are vital.

In this task, the Sony VPL-ES2 is worthy of a mention. This was the only model that had motorised pegs that raised themselves when the unit was powered up. A user can also control the vertical tilt of the unit electronically from a menu or via a remote. The VPL-ES2 also had a

lens cover which slid away when the unit was powered up and went back to protecting the lens when the unit was switched off—some very cool features that make remote operation easy.

This unit was by far the best designed projector of all, in terms of set up and adjustment, and placement connectors (which were on the right hand side). The knobs to adjust focus and zoom were also neatly placed on the right hand side, further easing setup.

While we are on the subject of ease of set up, we should also mention the Sharp PG-B10S. This was the only entry-level projector that had a "Lens Shift" feature. A tiny joystick beside the lens could be used to shift it vertically or horizontally. This helped us do away with the need to fidget with its pegs to get the image right.

Remote Control

Remote controllers are a common add-on to a projector. However, a few of the projectors had remotes that doubled up as presentation tools and laser pointers. The Acer PD113 had one such remote which unshackled us from returning to the laptop for a slide change. This is a greatly appreciated feature. Its pointer function further lends flexibility to highlight important figures and areas on the screen without having to walk to the screen and block the image.

Presets

A preset mode is a zero-hassle way of setting up a projector. No need to play with the colour, brightness and contrast settings—just choose a preset and you are very likely set. Except for the Canon unit, everyone had at least three presets to choose from, the most common being Standard, Presentation and Cinema. Some models such as the Sharp PG-B10S and HP vp6110 also had 'Game' and 'Living Room' modes.

Size

Compact and lighter units with a decent feature-set and performance will always be preferred over bulky units offering similar or even slightly better performance. The Acer PD113 was very compact, just 9.2 cm in height and weighing a mere two kg. Its lens is placed well inside a cavity with a big rubber lens cap, which not only protects it from dust, but also from shocks and mechanical jerks. The Epson EMP-S1H had the largest body frame, and weighed 3.2 kg. However, it sported an elegant design that was sorely missed in the other projectors.

How Did They Perform?

As mentioned before, we concentrated on evaluating the textual quality of the projectors' output. For this, we used PowerPoint presentations employing slides with black text on white backgrounds, coloured text on coloured backgrounds, and graphs and animations. Refer to the box "How We Tested and Awarded" for more information.

Performance While Displaying Text



Epson EMP-S1H